PRE-PUBLICATION SCIENTIFIC DUE DILIGENCE

# SIMUPHARM.IO

Prior-Art, Novelty Decomposition, Empirical Benchmark Datasets & Scientific Validation Intelligence Report

**PROJECT**
SimuPharm.io Virtual Analytical Laboratory

**AUDIT PHASE**
Phase 2 — Novelty, Prior-Art & Benchmarks

**AUDIT DATE**
October 4, 2026

**CODEBASE STATUS**
FROZEN (Read-Only Forensic Audit)

**TARGET JOURNAL**
Journal of Chemical Education (ACS)

**PUBLICATION FEE**
$0 Mandatory APC (Subscription Route)

**RECOMMENDED CONCEPT**
Concept B: Fault Injection & Troubleshooting

**SCIENTIFIC STATUS**
HIGH NOVELTY in Fault Simulation; AI Purged

## EXECUTIVE AUDIT SUMMARY

SimuPharm possesses **genuine, defensible scientific and educational novelty** in its coupled hydrodynamic-chromatographic operational fault-injection engine (bubbles, leaks, column aging) for USP <621> diagnostic education. However, claims of "AI/ML", "Bayesian Knowledge Tracing", "Digital Twin", and "300+ compounds" are factually contradicted by codebase inspection and must be completely purged from all manuscripts. Publication should target the **Journal of Chemical Education** ($0 mandatory APC) under Concept B.

# 1. SimuPharm Stated Claims vs. Codebase Reality

An exhaustive forensic audit of SimuPharm's documentation, pitch decks, comments, and JavaScript/Python source code was performed to verify scientific claims against empirical reality. The results are summarized below.

| ID | STATED CLAIM | CODEBASE REALITY | AUDIT STATUS | ACTION |
|---|---|---|---|---|
| **CLM-01** | Mechanistic HPLC simulation using Linear Solvent Strength (LSS) | Implemented in `hplc_simulator_script.js` (lines 1800-1950): ln k = ln k0 - S*phi with Henderson-Hasselbalch ionization. | VERIFIED | Safe to publish; cite Snyder & Dolan. |
| **CLM-02** | Real-time numerical gradient integration (dt = 0.01 min) | Discrete time-stepping evaluates migration integral client-side in <5 ms without server dependencies. | VERIFIED | Safe to publish; document in Methods. |
| **CLM-03** | Fluidic pressure modeling via Kozeny-Carman & Grunberg-Nissan | Calculates Delta P = (180*eta*L*u)/(dp^2) with composition-dependent viscosity function. | VERIFIED | Safe to publish; standard physics. |
| **CLM-04** | Coupled operational fault injection (bubbles, leaks, aging) | Dynamic functions modify effective flow, plate counts, and inject stochastic baseline pulses. | VERIFIED | **CORE NOVELTY OF PAPER.** |
| **CLM-05** | Automated USP <621> system suitability calculation | Automated calculation of Rs, N, and tailing factor Tf according to pharmacopoeial formulas. | VERIFIED | Safe to publish; regulatory utility. |
| **CLM-06** | Extensive 300+ compound library | `backend/compounds/database.py` contains exactly **21 compounds**. | CONTRADICTED | **PURGE.** State exact count: 21 compounds. |
| **CLM-07** | AI & Machine Learning analytics | Rule-based window counting for peaks; cosine similarity is dead code; zero trained ML models. | CONTRADICTED | **PURGE.** Remove all AI/ML claims. |
| **CLM-08** | Bayesian Knowledge Tracing (BKT) | `adaptive_engine.py` is simple 1-line exponential moving average; zero BKT math. | CONTRADICTED | **PURGE.** State simple progress tracking. |
| **CLM-09** | Digital Twin of analytical instruments | Simulation runs entirely disconnected from physical instrument sensors; no IoT telemetry. | MISMATCH | **PURGE.** Rebrand as In-Silico Simulator. |
| **CLM-10** | World's first browser-based HPLC simulator | Boswell et al. published `hplc-simulator.org` in JCE in 2013; Perri published in 2020. | CONTRADICTED | **PURGE.** Acknowledge Boswell et al. |

# 2. Prior-Art Landscape & Master Matrix Analysis

SimuPharm exists within a rich 45-year history of separation science software. Existing systems broadly bifurcate into industrial optimization tools ($>\$10,000$, non-educational) and academic/procedural teaching tools (free, but idealized or non-mathematical).

| PLATFORM | TYPE & VENUE | UNDERLYING MATH | FAULT SIMULATION? | MULTI-INSTRUMENT? | NOVELTY THREAT TO SIMUPHARM |
|---|---|---|---|---|---|
| **Boswell et al. (2013)** HPLC-Simulator.org | Academic Web Tool *J. Chem. Educ.* (ACS) | Mechanistic LSS + Knox plate height + Darcy pressure | NO (0%) | NO (HPLC ONLY) | HIGH on basic LSS simulation; **ZERO on fault injection & diagnostics.** |
| **Guillarme et al. (2012)** Practical HPLC | Academic Excel Tool Univ. of Geneva | Mechanistic LSS + Kinetic plots + UHPLC pressure | NO (0%) | NO (HPLC ONLY) | LOW. Desktop Excel dependency; no web interactivity; no fault injection. |
| **Perri (2020)** ChemCompute HPLC | Academic Jupyter *J. Chem. Educ.* (ACS) | Empirical polynomial data generator | NO (0%) | NO (HPLC ONLY) | LOW. Server-side Jupyter execution; no real-time canvas UI; no faults. |
| **DryLab 4** Molnár-Institute | Commercial Desktop Industrial QbD Standard | Multi-factorial 3D LSS design spaces (tG x T x pH) | NO (0%) | NO (HPLC ONLY) | ZERO. $>\$10,000$ license; requires real experimental runs; not educational. |
| **Labster HPLC Lab** Labster ApS | Commercial EdTech WebGL 3D Gamification | **NONE (Static pre-rendered animations)** | PROCEDURAL ONLY | YES (3D ROOMS) | ZERO scientifically. Purely scripted click-through game; no physical ODEs. |
| **Virtual Labs India** Ministry of Education / IITs | Government Open National Web Platform | Static lookup tables & pre-stored images | NO (0%) | YES (BROAD SCIENCE) | LOW. Outdated technology; no dynamic numerical gradient solvers. |

# 3. Head-to-Head Comparison: Boswell et al. (2013) vs. SimuPharm

Because Boswell et al. is the primary academic reference, a detailed feature comparison establishes SimuPharm's defensible scientific advance.

### BOSWELL ET AL. (JCE 2013)

- **Primary Focus:** Method development & parameter optimization.
- **Instrument Assumption:** 100% pristine, idealized hardware.
- **Fault Simulation:** Completely absent (zero failure modes).
- **Peak Synthesis:** Ideal Gaussian or basic convolution.
- **Scope:** Standalone HPLC simulator only.
- **Pedagogy:** Finding optimal %B, flow rate, and column dimensions.

### SIMUPHARM.IO (2026 AUDIT)

- **Primary Focus:** Operational troubleshooting & diagnostic literacy.
- **Instrument Assumption:** Dual mode: nominal vs. injected malfunctions.
- **Fault Simulation:** Air bubbles, pump leaks, column aging, cavitation.
- **Peak Synthesis:** Exponentially Modified Gaussian (EMG) via rational Chebyshev.
- **Scope:** Unified 8-instrument pharmaceutical analytical laboratory.
- **Pedagogy:** Diagnostic reasoning under USP <621> system suitability rules.

**The Fundamental Defensible Difference**

Boswell et al. built a simulator to show students how to optimize a separation when an instrument works perfectly. SimuPharm builds upon that foundation to teach students what to do when the instrument *fails*—bridging a major documented pedagogical void in pharmaceutical quality control education without risking \$50,000 hardware or toxic solvent waste.

## 4. Novelty Decomposition (N01 to N18)

SimuPharm's technical components were systematically classified into standardized novelty tiers to prevent over-claiming and ensure bulletproof peer review defense.

| ID | COMPONENT | NOVELTY TIER | AUDITOR JUSTIFICATION | DEFENSIBILITY |
|---|---|---|---|---|
| **N01** | Browser-Native Architecture | KNOWN | Boswell (2013), Perri (2020), and PhET (2012) run natively in browsers. | 10 / 10 |
| **N02** | Mechanistic HPLC Simulation | KNOWN | LSS and Van Deemter theory are 40-year-old established analytical chemistry. | 10 / 10 |
| **N03** | Numerical Gradient Integration | KNOWN | Discrete time-stepping (dt=0.01 min) is standard numerical ODE stepping. | 10 / 10 |
| **N04** | LSS Retention Modeling | KNOWN | Standard textbook Snyder-Dolan logarithmic solvent strength formulation. | 9 / 10 |
| **N05** | Van Deemter Band Broadening | KNOWN | H = A + B/u + C*u flow-rate dispersion is classical separation theory. | 10 / 10 |
| **N06** | EMG Asymmetric Peak Synthesis | COMBINED WELL | EMG is known; fast rational Chebyshev erfc(z) algorithm achieves 60fps web rendering. | 9 / 10 |
| **N07** | Kozeny-Carman Pressure Model | KNOWN | Standard porous media fluid mechanics coupled with Grunberg-Nissan viscosity. | 10 / 10 |
| **N08** | USP <621> System Suitability | KNOWN | Mathematical definitions standardized by pharmacopoeias; novelty is educational. | 10 / 10 |
| **N09** | **Systemic Fault Injection** | NOVEL | **First open web simulator coupling physical hydrodynamic-chromatographic faults.** | **9 / 10** |
| **N10** | **Troubleshooting Pedagogy** | NOVEL | **Pioneers diagnostic QC failure analysis in an interactive simulation sandbox.** | **9 / 10** |
| **N11** | Multi-Instrument Suite | PARTIALLY NOVEL | Lightweight client-side integration of 8 instruments without server compute. | 8 / 10 |
| **N12** | Unified QC Monograph Workflow | PARTIALLY NOVEL | Pedagogically novel in linking dissolution, UV assay, and HPLC purity testing. | 8 / 10 |
| **N13** | Adaptive Learning Engine | UNSUPPORTED | Code is simple EMA score updater; zero Bayesian Knowledge Tracing. | 2 / 10 |
| **N14** | AI / Machine Learning | UNSUPPORTED | Codebase contains zero trained ML models; peak detection is rule-based. | 1 / 10 |
| **N15** | Digital Twin | NOT DEFENSIBLE | Zero hardware sensor telemetry; violates ISO/IEEE 23247 standard. | 2 / 10 |
| **N16** | Compound / API Library | KNOWN | Curated set of 21 compounds; pitch claim of 300+ is factually false. | 7 / 10 |

| ID | COMPONENT | NOVELTY TIER | AUDITOR JUSTIFICATION | DEFENSIBILITY |
|---|---|---|---|---|
| **N17** | Automated Peak Integration | KNOWN | Standard baseline noise thresholding and trapezoidal integration. | 10 / 10 |
| **N18** | Zero-Latency Parameter Sweeps | COMBINED WELL | Sub-5ms update loop eliminates server roundtrip latency for responsive sliders. | 10 / 10 |

# 5. Empirical Benchmark Datasets for Physical Validation

SimuPharm's mathematical predictions must be quantitatively validated against authentic, published, peer-reviewed experimental datasets with fully defined, reconstructible chromatographic conditions.

| DATASET ID | CITATION & SOURCE | EXPERIMENTAL CONDITIONS | ANALYTES & METRICS | USEFULNESS SCORE |
|---|---|---|---|---|
| **DS-01** (Primary) | Tanaka et al. (1989) *J. Chromatogr. Sci.* 27, 721 DOI: `10.1093/chromsci/27.12.721` | C18 ($150 imes 4.6$ mm, $5\,\mu ext{m}$) Methanol/Water (80:20 & 70:30 v/v) 1.0 mL/min, $T = 30^\circ ext{C}$ & $40^\circ ext{C}$ | Uracil, Acetophenone, Benzene, Toluene, Naphthalene. Tabulated $t_R, k, lpha( ext{CH}_2), N, \Delta P$. | **10 / 10** (C18 Gold Standard) |
| **DS-02** (Secondary) | Franeta et al. (2002) *J. Pharm. Biomed. Anal.* 27, 1087 Waters App Note 720003922EN | C18 ($150 imes 4.6$ mm, $5\,\mu ext{m}$) MP A: 0.1% H3PO4 (pH 2.2); MP B: ACN Gradient: 10% to 70% B in 10 min | Paracetamol, Caffeine, Aspirin, Ibuprofen. Gradient $t_R, w_{0.5}, R_s, T_f$. | **9.5 / 10** (Pharmaceutical API QC) |
| **DS-03** (Stress-Test) | Boswell et al. (2013 SI) *J. Chem. Educ.* 90, 198 Supporting Information Dataset | Zorbax Eclipse XDB-C18 ($150 imes 4.6$ mm) ACN/H2O & MeOH/H2O gradients Flow: 0.5–2.0 mL/min, $T = 25-60^\circ ext{C}$ | Alkylbenzenes ($C_1 - C_6$), Uracil, Phenol. Empirical retention tables ($>50$ runs). | **9.0 / 10** (Direct Peer Comparison) |
| **DS-04** (Theory) | Snyder & Dolan (2007) *High-Perf. Gradient Elution* John Wiley & Sons | C18 ($150 imes 4.6$ mm, $5\,\mu ext{m}$) Gradient steepness $b = 0.05, 0.1, 0.2, 0.4$ Flow: 1.0–2.0 mL/min, $T = 35^\circ ext{C}$ | Standard neutral and acidic test probes. Band broadening, gradient $t_R$, $G$ factor. | **8.5 / 10** (Theoretical LSS Limit) |
| **DS-05** (Faults) | Dolan, J.W. (1983–2014) *LCGC North America* 'LC Troubleshooting' series | Deliberate fault injection on HPLC: Air in check valve, pump leak, frit void | Model solutes under fault injection. Qualitative baseline, pressure, and split peaks. | **9.0 / 10** (Definitive Fault Reference) |
| **DS-06** (Spectroscopy) | NIST SRM 935a (2011) NIST Spec. Publ. 260-54 US Dept. of Commerce | 10.00 mm quartz cuvette; 0.001 M HClO4 Concentrations: 20–100 mg/kg | Potassium dichromate (K2Cr2O7). Certified absorbance at 235, 257, 313, 350 nm. | **8.5 / 10** (Photometric Standard) |

# 6. Statistical Equivalence & Comparison Plan

To establish rigorous scientific validity in the manuscript, the simulated chromatographic predictions will be tested against empirical benchmarks using pre-specified statistical metrics.

### POINT-ACCURACY & LINEARITY

- **MAPE (Mean Absolute % Error):** Threshold $\le 3.0\%$ for isocratic $t_R$; $\le 5.0\%$ for gradient $t_R$.
- **RMSE (Root Mean Square Error):** Threshold $\le 0.15$ min across 12-min gradient runs.
- **Coefficient of Determination ($R^2$):** Linearity threshold $R^2 \ge 0.995$ for retention correlation; $R^2 \ge 0.999$ for UV absorbance.

### EQUIVALENCE & AGREEMENT

- **TOST Equivalence Testing:** Two One-Sided Tests ($p < 0.05$) within regulatory equivalence bounds $ heta = \pm 5\%$.
- **Bland-Altman Agreement:** Mean bias $|□ar{d}| < 0.05$ min; 95% of residuals within limits of agreement ($\pm 0.20$ min).
- **System Suitability Repeatability:** Repeatability $\%RSD \le 1.0\%$ for $t_R$; $\%RSD \le 2.0\%$ for peak area (USP <621>).

## 7. Journal Matching & $0-APC Strategy

The user's explicit requirement is to publish in reputable peer-reviewed journals without paying mandatory author publication fees ($0 mandatory APC).

| JOURNAL NAME | PUBLISHER | TRADITIONAL ROUTE ($0 APC)? | ARTICLE TYPE | FIT SCORE /100 | STRATEGIC STATUS |
|---|---|---|---|---|---|
| **Journal of Chemical Education** | ACS | YES ($0 MANDATORY APC) | Technology Report | **95 / 100** | **PRIMARY TARGET (RANK 1)** |
| **Comput. Appl. in Eng. Education** | Wiley | YES ($0 MANDATORY APC) | Research Article | **88 / 100** | **SECONDARY TARGET (RANK 2)** |
| **Pharmacy Education** | FIP | YES (DIAMOND OA: $0) | Innovation Report | **84 / 100** | **TOP PHARMACY TARGET** |
| **J. Pharm. Biomed. Analysis** | Elsevier | YES ($0 MANDATORY APC) | Short Communication | **80 / 100** | Solid Analytical QC Venue |
| **Analytical Methods** | RSC | YES ($0 MANDATORY APC) | Technical Note | **76 / 100** | Reputable RSC Option |
| **SoftwareX** | Elsevier | NO ($950 MANDATORY APC) | Software Publication | **68 / 100** | Avoid (Mandatory Fee) |
| **Journal of Chromatography A** | Elsevier | YES ($0 MANDATORY APC) | Regular Paper | **62 / 100** | High Desk-Reject Risk |
| **Chemometrics & Intell. Lab. Syst.** | Elsevier | YES ($0 MANDATORY APC) | Research Paper | **50 / 100** | Fatal Risk (No True ML) |
| **Lab on a Chip** | RSC | YES ($0 MANDATORY APC) | Full Paper | **0 / 100** | **100% Scope Mismatch** |

# 8. Mandatory Red-Flag Claim Blacklist

**PROHIBITED CLAIMS — MUST NEVER APPEAR IN MANUSCRIPT**

- ❌ **DO NOT CLAIM "Artificial Intelligence" or "Machine Learning":** Codebase inspection confirms zero machine learning models. Claiming ML will guarantee instant hostile rejection.
- ❌ **DO NOT CLAIM "Bayesian Knowledge Tracing (BKT)":** The adaptive engine is a simple 1-line exponential moving average.
- ❌ **DO NOT CLAIM "Digital Twin":** SimuPharm lacks bidirectional real-time sensor telemetry with physical hardware (violates ISO/IEEE 23247).
- ❌ **DO NOT CLAIM "300+ Compound Database":** The code contains exactly 21 compounds. Claiming 300+ destroys author credibility.
- ❌ **DO NOT CLAIM "World's First Web HPLC Simulator":** Boswell et al. published `hplc-simulator.org` in JCE in 2013.
- ❌ **DO NOT CLAIM "Optical Fourier Transform Interferometry in FTIR":** Code directly synthesizes Gaussian bands in wavenumber space.
- ❌ **DO NOT CLAIM "Lab-on-a-Chip / Microfluidics":** SimuPharm models standard macro-bore analytical columns (4.6 mm ID), not microfluidics.

# 9. Executive Phase 2 Decision Memo

Below is the formal, consolidated Phase 2 Decision Memo summarizing all scientific due diligence conclusions for SimuPharm.io.

## PHASE 2 DECISION MEMO — SIMUPHARM.IO

| | |
|---|---|
| **STRONGEST CONTRIBUTION:** | Coupled hydrodynamic-chromatographic operational fault injection (bubbles, leaks, column degradation) within a real-time client-side numerical gradient engine for USP <621> diagnostic education. |
| **CURRENT NOVELTY LEVEL:** | **GENUINELY NOVEL in Fault Simulation & Diagnostic Pedagogy (N09, N10);** KNOWN in basic LSS and Van Deemter math (N02–N05); UNSUPPORTED in AI/ML (N13–N15). |
| **BEST VALIDATION DATASET:** | **Primary:** Tanaka Test Mixture on C18 (DS-01, Tanaka 1989 / Snyder 2002);<br>**Secondary:** 4-Component Pharmaceutical Analgesic Mixture (DS-02, Waters / Franeta 2002). |
| **MOST IMPORTANT EXPERIMENT:** | Execution of the 20-run validation protocol in SimuPharm comparing simulated $t_R, k, \Delta P, R_s$ against Tanaka (DS-01) and Analgesic (DS-02) empirical data. |
| **MOST DANGEROUS CLAIM:** | FATAL RISK Claiming "Artificial Intelligence", "Bayesian Knowledge Tracing", or "Digital Twin" when the code contains only simple heuristic rules. |
| **BEST PAPER CONCEPT:** | **CONCEPT B (Score: 94/100):** *SimuPharm: An Interactive In-Silico HPLC Troubleshooting and Fault-Injection Workbench for Pharmaceutical Quality Control Education.* |
| **BEST TARGET JOURNAL:** | **Journal of Chemical Education (ACS)** under Technology Report article type.<br>**Publication Fee:** 100% FREE ($0 mandatory APC on subscription route). |
| **MINIMUM REQUIRED EVIDENCE:** | 1. Empirical validation matrix showing MAPE < 3.0% against Tanaka/Analgesic data;<br>2. Dolan fault signature match;<br>3. Pilot student usability trial ($n = 25$, SUS > 75). |
| **BIGGEST REASON FOR REJECTION:** | If the authors over-claim AI/ML, present LSS equations as novel, ignore Boswell et al. (2013), or claim a 300+ compound library without proof. |
| **NEXT PHASE (PHASE 3):** | **EXECUTE BENCHMARK VALIDATION EXPERIMENTS:** Run the 20 test conditions in SimuPharm, compile statistical agreement tables, and draft the JCE manuscript. |

SimuPharm.io Scientific Due Diligence Dossier • Phase 2 Audit Completed • Frozen Codebase Status Confirmed